

High School Computer Science Assessment (Python)

Student Performance Task: Student Score Analyzer

Time allowed: 60 minutes

Programming language: Python

Focus: Lists, loops, conditionals, basic data handling

A school has recorded the test scores of students in a class. You are given two related lists: student names and their corresponding scores. Your task is to process this data to produce meaningful analysis.

IMPORTANT INSTRUCTIONS: - You must complete all tasks in Python. - For each task, you MUST also write pseudocode before your Python solution. - Pseudocode should explain your logic in plain steps. - Marks are awarded for logic, correctness, and clarity.

Starter Data:

```
students = ["Alice", "Ben", "Chloe", "Daniel", "Eva"]
```

```
scores = [78, 55, 92, 67, 81]
```

Each index corresponds between the two lists.

Task 1 (20 marks)

Display each student and score. Format: Name: Score. Pseudocode required.

Task 2 (15 marks)

Calculate and print average score (1 decimal place). Pseudocode required.

Task 3 (20 marks)

Find highest score and corresponding student. Pseudocode required.

Task 4 (20 marks)

Count how many students passed (≥ 60) and failed. Pseudocode required.

Task 5 (25 marks)

List students scoring above average. Pseudocode required.

Challenge (+10 marks)

Sort students by score (highest to lowest) and display ranking. Pseudocode required.

Marking Scheme (Total: 100 + 10 bonus)

Logic & pseudocode: 20

Loops & structure: 15

Data handling: 20

Correct calculations: 20

Output formatting: 10

Problem solving: 15

Extension: +10

Students should NOT use external libraries. Only built-in Python features are required. Focus on clarity and correct reasoning before coding.